VELS
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PALLAVARAM, THALAMBUR, PERIYAPALAYAM, THIRUVANMIYUR

SmartPark - loT and ML Based Real-Time Parking Slot Availability System for College Campus

ABSTRACT

Parking areas can be hard to use when drivers don't know which spots are open or when reserved spots will be free. This paper introduces SMART PARK, a system that brings together Internet of Things (technology with machine learning to predict space availability. An infrared (IR) sensor is attached to each parking spot. An ESP32 device sends updates via Wi-Fi to Firebase indicating whether the spot is taken. A mobile app shows the status of each spot and allows people to make reservations based on a start date, start time and end time. The app checks existing reservations before confirming one. The times when parking sessions start and end are also saved to create a history that can be looked at later. A Random Forest model uses information such as the day of the week, the time, the weather and past space usage to predict how many spots will be free at a given time. This system brings together checking, planning ahead, booking and predicting all in one process. Testing with a planned prototype is still needed, and possible improvements could include automatic number-plate recognition and more ways to handle reservations.

Key Words — IOT; Smart Parking; ESP32; IR Proximity Sensor; Cloud/Firebase; Real-time Occupancy; Time- Reservation; Parking Session Management; Machine Learning.

TEAM MEMBERS

1.P.Suganthen

2.K.Sathish Kumar

3.K.Chandru

PROJECT-GUIDE SIGNATURE

Dr.N.Meenakshi,

(Assistant Professor)